

OBSTACLE AND BOUNDARY AVOIDING ROBOT (OBAR)

A robot is an intelligent electromechanical system that senses its environment, processes information, and performs actions automatically.

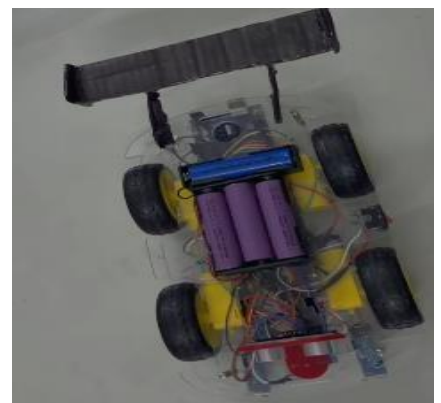
The **Obstacle Detection and Boundary (Black Line) Avoiding Robot** is an autonomous mobile robot designed to navigate safely by detecting obstacles using an ultrasonic sensor and identifying black boundary lines using IR sensors. It automatically changes its direction to avoid collisions and prevents crossing predefined restricted areas without requiring human intervention. Such robots improve safety, efficiency, and reliability in automated environments. Similar technologies are widely used in **warehouse robots** such as those operated by Amazon Robotics, **autonomous vacuum cleaners** such as iRobot Roomba, **hospital service robots** for medicine and supply delivery, and **automated guided vehicles (AGVs)** used in factories for material transportation.

CREATED BY:

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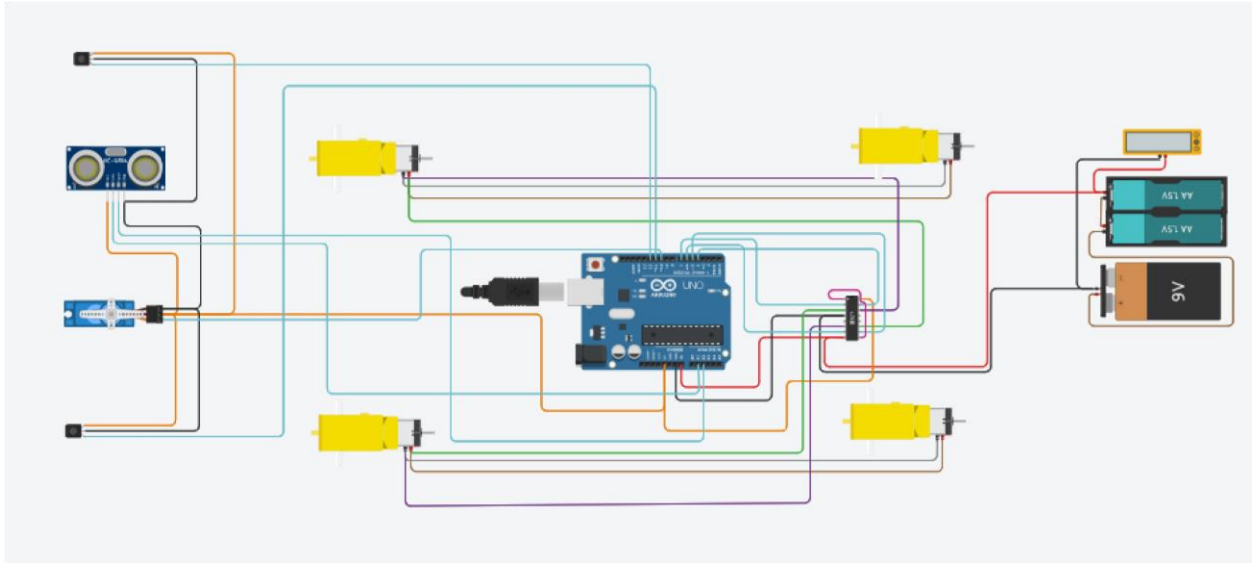
COMPONENTS USED

- ARDUINO UNO R3
- L28N MOTOR DRIVER
- 4WD ROBO KIT
- 4 DC GEAR MOTORS WITH WHEELS
- SERVO MOTOR
- IR SENSOR
- ULTRASONIC SENSOR
- JUMPER WIRES
- AA SIZE LITHIUM ION BATTERY 4X
- BATTERY HOLDER
- SWITCH
- ULTRA SONIC SENSOR HOLDER
- SCREWS,NUTS,SPACER



SIMULATION

I used TinkerCad for circuit diagram. This software doesn't have the L28N motor driver so as an alternative I used L293D IC which is also a motor driver IC. [click here for diagram](#)

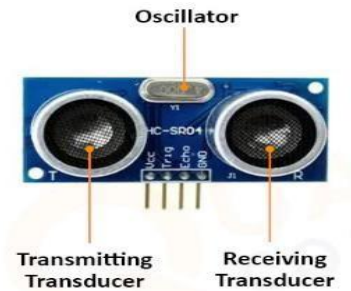


In this diagram

- Orange wire is for 5 volt
- Red for 12v
- Black for GND
- Turquoise for Arduino pins
- Green for + of motor
- Purple for – of motor
- Brown and Grey for making common connection

Ultrasonic SENSOR (HC-SR04)

The **HC-SR04 Ultrasonic Sensor** is an electronic distance-measuring sensor used to detect objects without making physical contact. It operates by transmitting **ultrasonic sound waves**, which are sound waves with frequencies **greater than 20 kHz (20,000 Hz)**. The normal range of **human hearing is approximately 20 Hz to 20 kHz**, so the ultrasonic waves produced by this sensor cannot be heard by humans. The HC-SR04 specifically operates at a frequency of **40 kHz**, making it suitable for accurate distance measurement and obstacle detection in robotic applications.



The sensor contains an **internal oscillator** that generates a stable **40 kHz electrical signal**. This signal is converted into ultrasonic sound waves by the transmitting transducer. When the Arduino sends a **10-microsecond HIGH pulse** to the **TRIG (Trigger)** pin, the internal oscillator becomes active and causes the transmitting transducer to emit **eight consecutive ultrasonic pulses at 40 kHz**. These sound waves travel through the air until they strike an object and are reflected back toward the sensor.

The HC-SR04 has **four pins**. The **VCC** pin provides the **5V power supply** required to operate the sensor, while the **GND (Ground)** pin completes the electrical circuit. The **TRIG (Trigger)** pin is an input pin that receives a signal from the Arduino to start a measurement. The **ECHO** pin is an output pin that goes HIGH after the ultrasonic pulse is transmitted and remains HIGH until the reflected echo is received. By measuring the duration of this HIGH pulse, the Arduino calculates the distance between the sensor and the object.

The two circular **speaker-shaped components** on the front of the HC-SR04 are called **ultrasonic transducers**. The **left transducer acts as the transmitter (TX)** and converts electrical signals generated by the oscillator into **40 kHz ultrasonic sound waves**. The **right transducer acts as the receiver (RX)** and detects the reflected ultrasonic waves (echoes) returning from nearby objects. The sensor then sends the echo timing information to the Arduino, allowing it to determine the object's distance accurately. In an obstacle-avoiding robot, the HCSR04 functions as the robot's **eyes**, continuously detecting obstacles so that the robot can change its direction and avoid collisions.



SERVO MOTOR

A **servo motor** is a rotary actuator designed to provide **precise control of angular position**, rather than continuous rotation like a normal DC motor. It is widely used in robotics, automation, and control systems where accurate movement is required. A standard hobby servo can rotate to a specific angle, typically between **0° and 180°**, and hold that position until a new command is received. In an obstacle-avoiding robot, the servo motor is used to rotate the **ultrasonic sensor** so that the robot can scan its surroundings, detect obstacles on the left and right, and decide the safest direction to move.

A servo motor consists of four main parts: a **DC motor**, a **gearbox**, a **position sensor (potentiometer)**, and a **control circuit**. The DC motor provides the rotational force, while the

gearbox reduces the speed and increases the torque, enabling the servo to move accurately. The potentiometer continuously measures the current shaft position, and the control circuit compares this position with the desired position sent by the Arduino. If there is any difference, the control circuit rotates the motor until the shaft reaches the required angle and then stops the motor, allowing the servo to maintain its position accurately.

A standard servo motor has **three wires (pins)**. The **VCC** wire (usually red) is connected to a **5V power supply** to operate the servo. The **GND** wire (usually brown or black) is connected to the Arduino ground to complete the electrical circuit. The **Signal** wire (usually orange, yellow, or white) is connected to one of the Arduino's digital pins. This signal wire receives **Pulse Width Modulation (PWM)** signals from the Arduino, which determine the desired angular position of the servo.

Unlike a DC motor, a servo motor is controlled by the **width of PWM pulses** rather than by continuously changing the voltage. The Arduino sends a pulse approximately every **20 milliseconds (50 Hz)**. The width of each pulse tells the servo which angle to move to. Typically, a pulse of about **1 ms** moves the servo close to **0°**, a **1.5 ms** pulse moves it to approximately **90°**, and a **2 ms** pulse moves it close to **180°**. The servo's internal control circuit interprets these pulse widths and rotates the shaft to the corresponding position.

The gearbox inside the servo plays an important role by **reducing the motor's speed and increasing its torque**, allowing the servo to move smoothly and accurately even when carrying a load. At the same time, the **potentiometer** continuously provides feedback about the shaft's position. This **closed-loop feedback system** ensures that the servo reaches the commanded angle precisely and holds that position until another command is received.

In this obstacle-avoiding robot, the servo motor rotates the ultrasonic sensor to different angles, such as left, center, and right. The Arduino measures the distance in each direction using the ultrasonic sensor and then determines the safest path for the robot. Therefore, the servo motor acts as the robot's **neck**, enabling the sensor to scan the environment instead of looking in only one direction.

Arduino UNO R3



The **Arduino Uno** is an open-source microcontroller development board based on the **ATmega328P** microcontroller. It is one of the most popular embedded system platforms used for electronics, robotics, automation, and Internet of Things (IoT) projects. The Arduino Uno serves as the **brain of the system**, executing the program uploaded by the user and controlling the connected sensors, motors, and other electronic components. It processes input signals from sensors, makes decisions according to the programmed logic, and generates output signals to control actuators.

The Arduino Uno operates at a **clock frequency of 16 MHz**, which means its microcontroller can execute **16 million clock cycles every second**. It has **32 KB of Flash memory** for storing the program, **2 KB of SRAM** for temporary data storage during execution, and **1 KB of EEPROM** for storing data permanently even when the power is turned off.

One of the key features of the Arduino Uno is its **14 digital input/output pins (D0–D13)**. These pins can be configured as either **INPUT** or **OUTPUT** using the `pinMode()` function. When used as inputs, they can receive digital signals (HIGH or LOW) from switches and sensors. When used as outputs, they send HIGH (5V) or LOW (0V) signals to control external devices such as LEDs, relays, and motor drivers.

Among the digital pins, **six pins (3, 5, 6, 9, 10, and 11)** are marked with the '~' symbol, indicating that they support **Pulse Width Modulation (PWM)**. These pins can generate PWM signals using the `analogWrite()` function. PWM rapidly switches the output between HIGH and LOW, producing an average output that behaves like different voltage levels. PWM is commonly used for **controlling the speed of DC motors, adjusting LED brightness, and driving other compatible devices**.

The Arduino Uno also contains **six analog input pins (A0–A5)**. Their primary purpose is to measure analog voltages between **0V and 5V** from sensors such as temperature sensors, light sensors (LDRs), gas sensors, and potentiometers. The Arduino uses a built-in **10-bit Analog-toDigital Converter (ADC)** to convert these analog voltages into digital values ranging from **0 to 1023**. Besides reading analog signals, these analog pins can also function as ordinary digital input/output pins whenever additional digital pins are required.

The board provides several **power pins** to operate both the Arduino and external components. The **5V pin** supplies a regulated 5V output for powering sensors and modules. The **3.3V pin** supplies 3.3V to low-voltage devices. The **GND (Ground)** pins provide the common reference point and complete the electrical circuit. The **VIN** pin is used to supply an external voltage (typically **7V–12V**) to power the Arduino. When connected through VIN or the DC barrel jack, the onboard voltage regulator converts the input into stable operating voltages for the board.

The Arduino Uno also includes dedicated communication pins. **Pins D0 (RX) and D1 (TX)** are used for **UART serial communication**, allowing the Arduino to exchange data with a computer or other microcontrollers. The board also supports **I²C communication** using **A4 (SDA) and A5 (SCL)**, and **SPI communication** using **D10 (SS), D11 (MOSI), D12 (MISO), and D13 (SCK)**, making it compatible with a wide variety of sensors, displays, and communication modules.

The **RESET** pin and reset button restart the execution of the uploaded program whenever required. The **USB Type-B port** is used to upload programs from the Arduino IDE and can also provide 5V power to the board. A **DC barrel jack** allows the Arduino to be powered from an external adapter or battery.

In this obstacle-avoiding robot, the Arduino Uno acts as the **central controller**. It sends a trigger pulse to the ultrasonic sensor, receives the echo signal, calculates the distance to nearby objects, rotates the servo motor to scan different directions, and controls the L293D motor driver to move

the DC motors forward, backward, left, or right. Thus, it coordinates all the hardware components and ensures the robot operates intelligently to avoid obstacles.

IR SENSOR LM393



The **LM393-based IR Sensor Module** used in robotics for **obstacle detection, detection, and object sensing**. It

infrared (IR) light, which has a wavelength of approximately **700 nm to 1 mm** and lies just beyond the visible spectrum. Since the **human eye can only detect visible light with wavelengths of approximately 380 nm to 750 nm**, infrared light cannot be seen by humans. The sensor determines the presence of an object by emitting infrared light and detecting the amount of light reflected back from nearby surfaces.

is an infrared sensor widely **line following, edge** operates by transmitting

The module consists of an **infrared LED (IR Transmitter)**, an **infrared photodiode (IR Receiver)**, an **LM393 voltage comparator IC**, a **sensitivity adjustment potentiometer**, and a **status LED**. The IR LED continuously emits infrared light toward the surface or object. When this light strikes an object, part of it is reflected back to the photodiode. The amount of reflected light depends on the distance, color, and surface characteristics of the object. The photodiode converts the received infrared light into a small electrical signal.

The **LM393 comparator IC** is the main component of the module. Its purpose is to compare the signal produced by the photodiode with a **reference voltage** set using the onboard potentiometer. If the reflected infrared light is greater than the reference level, the LM393 changes the digital output state. This conversion from a small analog signal into a clean **HIGH or LOW digital signal** makes it easy for the Arduino to detect objects or lines without performing analog signal processing.

The **potentiometer** on the sensor module allows the user to adjust the **detection sensitivity**. Rotating the potentiometer changes the reference voltage of the LM393 comparator, thereby increasing or decreasing the detection range. This enables the sensor to be calibrated for different lighting conditions and different surface colors.

The module generally has **four pins: VCC, GND, DO (Digital Output), and AO (Analog Output)**. The **VCC** pin is connected to the **5V supply** from the Arduino, while the **GND** pin completes the electrical circuit. The **DO (Digital Output)** pin provides a HIGH or LOW signal after processing by the LM393 comparator and is commonly used in Arduino projects because it is simple to interface. Some versions of the module also provide an **AO (Analog Output)** pin, which outputs the raw analog voltage from the photodiode before comparison, allowing the Arduino to measure the intensity of reflected infrared light using an analog input.

In an obstacle-avoiding or line-following robot, the **DO pin** is usually connected to one of the

Arduino's digital input pins. The Arduino continuously reads the sensor output using the `digitalRead()` function. When the sensor detects an object or a black line (depending on the module and its calibration), the output changes state, allowing the Arduino to control the motors accordingly.

The LM393 sensor detects differences in **surface reflectivity**. **White or light-colored surfaces** reflect more infrared light, whereas **black or dark-colored surfaces** absorb most of the infrared light and reflect very little. This property allows the sensor to distinguish between black and white surfaces, making it ideal for line-following robots.

In your project, the LM393 IR sensor is mounted underneath the robot to detect the **black boundary line**. The Arduino continuously monitors the sensor output. When the sensor detects the black line (according to your module's logic level), the Arduino immediately changes the robot's direction, preventing it from crossing the boundary. Thus, the IR sensor enhances the robot's navigation and safety by enabling reliable line detection.



TT DC Gear Motor

The **TT DC Gear Motor** is a compact electric motor that combines a **DC motor** with a **gearbox** to produce **high torque at low speed**. Unlike a standard DC motor, which rotates at very high speed with relatively low torque, the TT gear motor uses a series of reduction gears to decrease the rotational speed while significantly increasing the output torque. This makes it ideal for mobile robots, robotic cars, and automation projects where controlled movement and sufficient pulling force are required. TT gear motors are commonly available in **3V, 6V, and 12V** versions, with the **3V–6V models** being the most widely used in educational robotics.

A TT gear motor consists of a **DC motor**, a **plastic reduction gearbox**, an **output shaft**, and **two power terminals**. When electrical power is supplied to the motor terminals, the DC motor begins to rotate at high speed. This rotation passes through multiple gears inside the gearbox, which reduce the speed and increase the torque before transferring the motion to the output shaft. The direction of rotation depends on the polarity of the supplied voltage. Reversing the polarity causes the motor to rotate in the opposite direction, allowing the robot to move both forward and backward.

In this obstacle-avoiding robot, the TT gear motors are responsible for converting electrical energy from the battery into mechanical motion. The Arduino controls the motors through the **L298N/L293D motor driver**, which allows the robot to move **forward, backward, turn left, and turn right** according to the distance measured by the ultrasonic sensor and the signals received from the IR sensors.

Although the robot has **four wheels**, only the **front two wheels are directly driven by two TT gear motors**, while the rear wheels rotate freely along with the movement of the robot. This design was intentionally chosen to **reduce the electrical load on the motor driver and battery**.

The L298N motor driver can safely supply a limited amount of current (typically around **2 A total** under normal operating conditions). If four motors were driven simultaneously, the current demand would increase significantly, especially when the battery voltage decreases or when the robot encounters additional load such as rough surfaces or obstacles. Under such conditions, a single TT motor may draw **approximately 1–1.5 A during startup or heavy load**, and four motors together could exceed the motor driver's current capability. Excessive current results in **voltage drop, overheating of the motor driver, reduced efficiency, and possible damage to the driver IC**. Therefore, using only two powered motors reduces current consumption, improves reliability, minimizes heating, and extends battery operating time while still providing sufficient traction for the robot.



L298N Motor Driver Module

The **L298N Motor Driver Module** is a dual H-Bridge motor driver used to control the **direction and speed of DC motors and stepper motors**. Since the Arduino Uno can provide only a small current of approximately **20–40 mA per I/O pin**, it cannot drive DC motors directly because motors require much higher current. The L298N acts as an interface between the Arduino and the motors by receiving low-current control signals from the Arduino and supplying the higher current and voltage required by the motors. It can independently control **two DC motors**, enabling the robot to move forward, backward, turn left, and turn right.

The main component of the module is the **L298N integrated circuit (IC)**, which contains **two H-Bridge circuits**. An **H-Bridge** is an electronic circuit that allows the polarity of the voltage applied to a DC motor to be reversed. By changing the polarity, the motor can rotate in either the clockwise or anticlockwise direction. The Arduino controls these H-Bridges through the input pins, enabling precise control of the robot's movement.

The L298N module supports a **motor supply voltage of 5V to 35V** and uses a separate **5V logic supply** to operate its internal circuitry. The module has a **12V (VS)** pin for connecting the motor power supply, a **5V** pin for the logic supply (or regulated 5V output when the onboard regulator is enabled), and a **GND** pin that provides the common ground connection between the Arduino, battery, and motor driver.

The module contains **four input pins (IN1, IN2, IN3, and IN4)** that receive digital signals from the Arduino. **IN1 and IN2** control **Motor A**, while **IN3 and IN4** control **Motor B**. Depending on the HIGH and LOW combinations applied to these pins, the motor driver determines the direction of rotation of each motor.

The **ENA** and **ENB** pins are the **Enable pins** for Motor A and Motor B. These pins enable or disable the corresponding motor channels. When connected to **5V (HIGH)**, the motors remain enabled continuously. If connected to the Arduino's **PWM pins**, they can receive PWM signals to control motor speed by varying the duty cycle. In this project, **ENA and ENB were connected directly to the Arduino's 5V pin** because speed control was not required. Therefore,

both motors remained permanently enabled, and only the direction was controlled through the input pins.

The **OUT1** and **OUT2** terminals are connected to **Motor A**, while **OUT3** and **OUT4** are connected to **Motor B**. These output terminals supply the required voltage and current to the motors according to the control signals received from the Arduino.

One of the important characteristics of the L298N is its **current handling capability**. The module can supply **up to 2 A peak current per channel**, while the **safe continuous current is typically around 1–1.5 A per channel**, depending on cooling conditions. During operation, if the motors draw excessive current for a long period, the L298N generates heat due to power dissipation. Excessive heating reduces efficiency and may activate thermal protection or even damage the motor driver if adequate cooling is not provided. Therefore, the L298N is best suited for **small and medium-sized DC motors**.

In this obstacle-avoiding robot, although the chassis has **four wheels**, only the **front two TT gear motors are powered**. This design choice was made to reduce the electrical load on the L298N motor driver. When DC motors start rotating or experience heavy mechanical load, they draw a **high starting (stall) current**, which can reach approximately **1–1.5 A per TT gear motor**. If all four motors were powered simultaneously, the total current demand could exceed the motor driver's safe operating limit, especially when the battery voltage decreases or the robot moves on rough surfaces. This could result in **overheating, voltage drop, reduced motor performance, and possible damage to the driver**. By powering only two motors, the current requirement is reduced, the L298N operates more efficiently, heating is minimized, battery life is improved, and the robot maintains reliable performance while still achieving effective movement.

In this project, the Arduino continuously receives information from the ultrasonic and IR sensors and sends control signals to the **IN1–IN4** pins of the L298N. The motor driver then supplies the required power to the TT gear motors, allowing the robot to move forward, reverse, turn left, or turn right according to the detected obstacles. Thus, the L298N serves as the **power interface between the Arduino and the DC motors**, enabling safe and efficient motor control.

Lithium-Ion Battery Pack



The **Lithium-Ion (Li-ion) battery** is a rechargeable energy source used to provide electrical power to the robot. It converts **chemical energy into electrical energy** through electrochemical reactions. Lithium-ion batteries are widely used in robotics because they offer **high energy density, lightweight construction, long cycle life, low self-discharge, and the ability to deliver high current** when required. These characteristics make them an excellent choice for powering motors, sensors, and microcontrollers in portable robotic applications. Each lithium-ion cell has a **nominal voltage of 3.7V** and reaches approximately **4.2V when fully charged**. A battery consists of two terminals: the **positive (+) terminal**, which supplies electrical

power to the circuit, and the **negative (-) terminal**, which completes the electrical circuit. When connected to the robot, electrons flow through the external circuit, supplying power to the Arduino, motor driver, and motors.

In this project, **four lithium-ion cells are connected in series** to increase the supply voltage. In a **series connection**, the **voltages of all cells add together**, while the **battery capacity (mAh)** remains the same as that of a single cell. Since each battery has a nominal voltage of **3.7V**, the battery pack provides a total nominal voltage of **14.8V ($3.7V \times 4$)**. When fully charged, each cell reaches **4.2V**, so the battery pack provides approximately **16.8V ($4.2V \times 4$)**. If each battery has a capacity of **1200 mAh**, the complete battery pack also has a capacity of **1200 mAh**.

The reason for using **four batteries instead of three** is to ensure a **stable and reliable power supply** throughout the robot's operation. As lithium-ion batteries discharge during use, their output voltage gradually decreases. This reduction in voltage can lower the speed and torque of the motors, reducing the overall performance of the robot. By starting with a higher supply voltage of **14.8V**, the robot continues to receive sufficient voltage even after the batteries begin to discharge. Since the **L298N motor driver supports motor supply voltages from 5V to 35V**, a **14.8V battery pack (16.8V when fully charged)** is well within its safe operating range. Therefore, the higher initial voltage compensates for the natural voltage drop during battery discharge, allowing the motors to operate efficiently for a longer period. As a result, the robot maintains **better speed, higher torque, improved stability, and more reliable obstacle avoidance performance** throughout its operating time.

The battery pack supplies power to the **L298N motor driver**, which then distributes the required power to the TT gear motors. The Arduino receives its operating voltage through an appropriate regulated supply, while the sensors and servo motor receive a stable **5V**. Thus, the lithium-ion battery pack serves as the primary power source of the robot and ensures continuous, stable, and efficient operation.

PSUEDO CODE

START

Set up:

Left motor

Right motor

Ultrasonic sensor

Left IR sensor

Right IR sensor

Servo motor

Point the servo straight ahead

Wait for 2 seconds

REPEAT FOREVER:

Check the left IR sensor

Check the right IR sensor

Check the ultrasonic sensor

IF an obstacle is 40 cm or closer

OR a black line is detected:

Stop the robot

Wait a little

Move backward

Wait a little

Stop the robot

IF the robot has tried to avoid obstacles 3 times:

Move backward farther

Stop

Choose LEFT or RIGHT randomly

Turn in the chosen direction

for a longer time

Reset the avoidance counter

ELSE:

Turn the servo to the RIGHT

Measure the distance on the RIGHT

Turn the servo to the LEFT

Measure the distance on the LEFT

Point the servo straight again

IF the RIGHT side has more space:

Turn RIGHT

ELSE:

Turn LEFT

Check the ultrasonic sensor again

ELSE:

Move FORWARD

Reset the avoidance counter

END REPEAT

END

Robot moves forward



Check obstacle + black line



Is danger detected?



NO YES



Move forward Stop



Move backward



Look RIGHT and LEFT



Which side is clearer?



RIGHT LEFT



Turn RIGHT Turn LEFT



Move again

The future belongs to those who innovate today. Robotics is not merely about building machines; it is about creating intelligent solutions that improve lives.